

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

9189/2

PAPER 2 Theory

NOVEMBER 2014 SESSION

1 hour 15 minutes

Candidates answer on the question paper.

Additional materials:

Data Booklet

Mathematical Tables and/or Calculator

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
TOTAL	

This question paper consists of 8 printed pages.

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- 1 A mixture of nitrogen and hydrogen was allowed to come to an equilibrium at 400 K. The initial amounts of nitrogen and hydrogen were 2.0 moles. At equilibrium 0.20 moles of nitrogen had reacted.

(a) Write an expression for K_p .

[1]

(b) Calculate the number of moles of $N_{2(g)}$, $H_{2(g)}$ and $NH_{3(g)}$ at equilibrium.

N_2

H_2

NH_3

[3]

(c) Given that the equilibrium constant at 400 K is 40.7 atm^{-2} , calculate the total pressure.

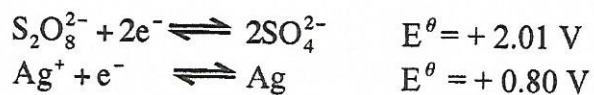
[3]

[Total: 7]

- 2 (a) Define the term *standard electrode potential*.

[1]

- (b) The standard electrode potentials for two half cells are given below



- (i) State the **three** conditions for which the term *standard* refers.

1.

2.

3.

- (ii) Write down a chemical equation for the overall reaction of the cell.

- (iii) Calculate the cell potential, E^θ cell, for the reaction.

- (iv) Sketch a diagram of the peroxodisulphate half cell.

[7]

- (c) State and explain, the effect, on the magnitude of E^θ cell of adding concentrated ammonia to the Ag^+/Ag half cell.

effect:

explanation:

[3]
[Total: 11]

- 3 Sulphur dioxide is a poisonous gas with adverse environmental effects.

- (a) (i) Give any **one** source of sulphur dioxide.

- (ii) State any **two** adverse effects of sulphur dioxide on the environment.

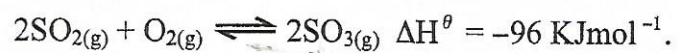
1.

2.

- (iii) Explain how sulphur dioxide preserves food.

[4]

- (b) Sulphur dioxide can be converted to sulphuric acid by the contact process. The reaction is



Give reasons for

- (i) cooling the reaction mixture,

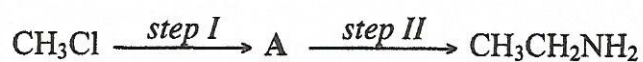
- (ii) using a low pressure even though high pressure produces a high yield,

- (iii) pumping an excess of air,

- (iv) using vanadium (V) oxide.

[4]
[Total: 8]

- 4 (a) Ethylamine can be prepared from chloromethane as shown.



- (i) State the reagents and conditions for *steps I* and *II*.

step I: reagent _____

condition(s) _____

step II: reagent _____

condition(s) _____

- (ii) Draw the displayed structural formula of the intermediate A.

[5]

- (b) (i) Dimethylamine and ethylamine are isomers. Give the structural formulae of the two isomers.

- (ii) Explain why dimethylamine is more basic than ethylamine.

[3]

For
Examiner's
Use

- (c) (i) State any **two** industrial uses of halogenated hydrocarbons.

1. _____
2. _____

- (ii) Describe the environmental consequences of the uncontrolled use of halogenated compounds.

[4]

[Total: 12]

- 5 **Fig.5** shows the structure of fluorescamine, a reagent for the detection of primary amines.

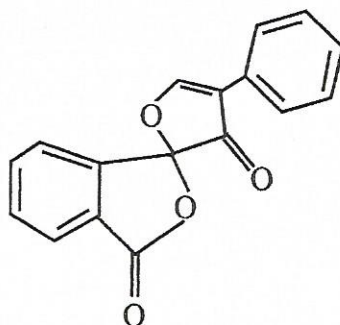


Fig.5

- (a) Identify any **two** functional groups present in this compound.

1. _____
2. _____

[2]

- (b) Draw the structure of the organic product formed when fluorescamine reacts with

- (i) NaOH under reflux,

(ii) NaBH_4 ,

(iii) Br_2 in the presence of FeBr_3 ,

(iv) hot concentrated KMnO_4 .

[4]

(c) Suggest explanations for the following observations:

(i) hexanedioic acid and 1,6-diaminohexane are formed when sulphuric acid is added to a fabric made of nylon-6,6

(ii) terylene is a better sweat absorber than nylon-6,6,

(iii) commercial fertilisers use urea, NH_2CONH_2 , to promote plant growth

[4]

[Total: 10]